# Inverted MAML Card 1: Round 5 Thermal Bit-Slicing Report

**Hardware:** Card 1 (T120+T20, 1k/2.2k) | **FW:** v2.2-MAML (0x20261030) | **Params:** 40% PWM, K=1, LR=0.03, L2=1024, Mom=0.85, Clamp=±75 LSB

## 1. Round 5 Summary & Cumulative Comparison (R1–R5)

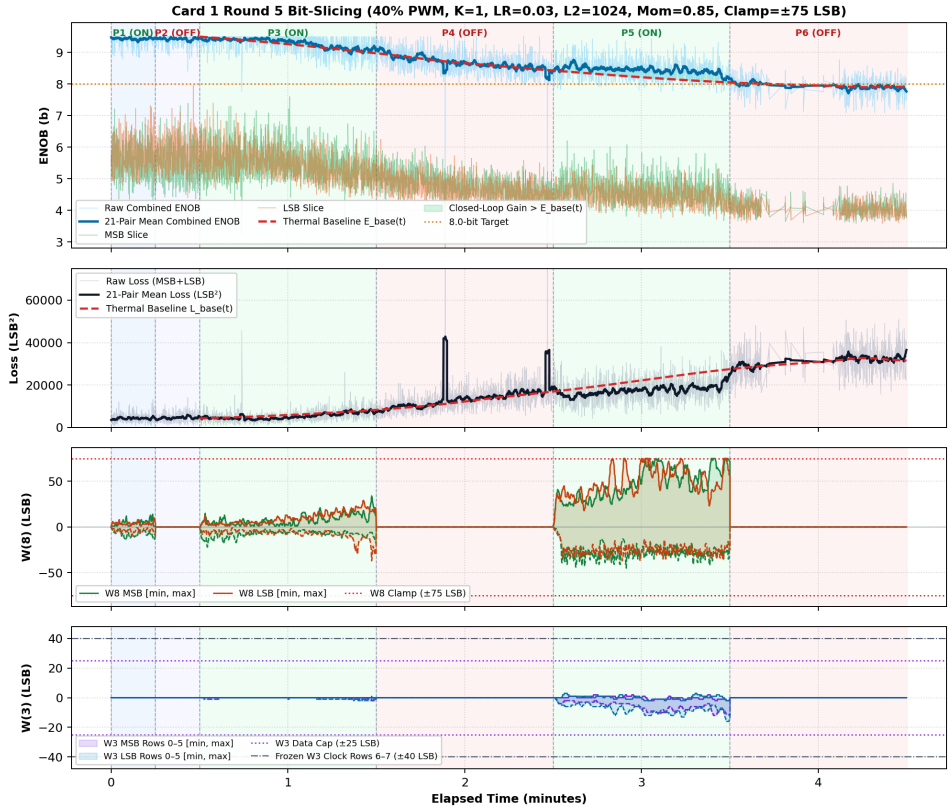
• **Cold Verification:** Phase 1 ON = **9.45 b**, Phase 2 OFF = **9.45 b** (W8=0 strictly).

• **Round 5 Detrended Trend Metrics:** Lift vs E_base = **+0.20 b**, Loss Ratio = **1.34x**, Climb = **+0.43 b**, Jitter = **0.26 b**, Recovery = **0.89**, Score = **+1.147**.

| Rnd | LR | L2 | Mom | Clamp | Lift | LossR | Climb | Jitt | Recov | Score |
|---|---|---|---|---|---|---|---|---|---|---|
| R1 | 0.03 | 1024 | 0.9 | ±75 | +0.66b | 1.57x | +0.74b | 0.33b | 0.67 | +1.957 |
| R2* | 0.035 | 1536 | 0.85 | ±75 | +0.50b | 1.38x | +0.82b | 0.29b | 0.96 | +2.004 |
| R3 | 0.02 | 2560 | 0.8 | ±70 | +0.09b | 1.11x | +0.22b | 0.26b | 1.00 | +0.559 |
| R4 | 0.045 | 2048 | 0.88 | ±75 | +0.11b | 1.17x | +0.14b | 0.24b | 1.00 | +0.643 |
| R5 | 0.03 | 1024 | 0.85 | ±75 | +0.20b | 1.34x | +0.43b | 0.26b | 0.89 | +1.147 |

## 2. Phase-by-Phase Telemetry (3,901 Pairs)

| Phase | Regime | N | Start->Late (Mean) | Δ vs E_base | Loss Start->Late | W8 Pk [M,L] | W3 Pk | W8 nz |
|---|---|---|---|---|---|---|---|---|
| P1 | Cold / Regler ON | 321 | 9.45->9.44 (9.43b) | - | 3635->4235 | [14,10] | 0 | 320 |
| P2 | Cold / Regler OFF | 303 | 9.45->9.42 (9.43b) | - | 3554->4621 | [0,0] | 0 | 0 |
| P3 | Warm / Regler ON | 1037 | 9.46->9.13 (9.30b) | +0.05 b | 4352->6638 | [34,37] | 2 | 1037 |
| P4 | Warm / Regler OFF | 927 | 9.04->8.52 (8.68b) | +0.01 b | 7725->17229 | [0,0] | 0 | 0 |
| P5 | Warm / Regler ON | 834 | 8.49->8.38 (8.45b) | +0.28 b | 18127->18807 | [75,75] | 16 | 834 |
| P6 | Warm / Regler OFF | 479 | 8.04->7.89 (7.93b) | -0.02 b | 28910->32304 | [0,0] | 0 | 0 |



## 3. All W(8) (7x6 MSB+LSB) & W(3) (8x6 MSB+LSB) Weight Dynamics — Round 5

• **Full 180-Synapse Per-Pair Capture:** Every individual weight in **Matrix 8 MSB (42 weights)**, **Matrix 8 LSB (42 weights)**, **Matrix 3 MSB (36 active + 12 clock)**, and **Matrix 3 LSB (36 active + 12 clock)** is plotted below across all columns `Col 0..5`. W(8) enforces anti-windup clamping at ±75 LSB, while W(3) rows 0–5 stay within ±25 LSB so rows 6–7 clock staircases remain locked at ±40 LSB.

**Round 5 Full Hardware Weight Telemetry: ALL 84 W(8) [7x6 MSB+LSB] & ALL 96 W(3) [8x6 MSB+LSB] Synapses**

